

Medical Management of Ectopic Pregnancy Protocol (CG623)

Approval and Authorisation

Approved by	Job Title	Date
Gynaecology Clinical Governance	Chair, Gynaecology Clinical Governance	20 th June 2025

Change History

Version	Date	Author	Reason
3	Jan 2021	Dr Caroline Prentice ST7 O&G (Apyll Chase, Consultant Gynae)	Reviewed and updated in line with NICE and move to EPR for clinical records
4	March 2023	C Prentice, Consultant O&G A Chase, Consultant O&G	Reviewed and amended to reflect current practice and update process flowchart (pg 4/5).
5	February 2025	C Prentice, Consultant O&G S Bhagwat, ST7 Obs & Gynae	Biennial review – minor changes.

To be read in conjunction with:

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1.0 Purpose

This protocol was originally developed in line with the Joint Royal College of Obstetricians and Gynaecologists & AEPU: *Green Top Guideline No. 21: Diagnosis and Management of Ectopic Pregnancy* (2016).

The purpose of this protocol is to offer to carefully selected patients medical management of ectopic pregnancy by the administration of intramuscular Methotrexate.

2.0 Definition

Systemic Methotrexate (MTX) Treatment

Methotrexate is a folic acid-antagonist (anti-metabolite) which prevents the growth of rapidly dividing cells by interfering with DNA synthesis. It can be administered systemically (IV, IM or orally). However, it is most commonly given according to a single-dose protocol which involves a single intramuscular dose of 50 mg/m². A single injection of methotrexate is well tolerated and is effective. Published studies have shown a success rate varying from 65.5% to 94% for single dose administration.

3.0 Patient Selection Criteria

- No significant pain
- No intrauterine pregnancy
- Unruptured ectopic pregnancy (diagnosed with BHCG and TVS) – mass <35mm. No FH
- BHCG value <1500 IU/litre - consider conservative management (see Ectopic Pregnancy GL1061)
- BHCG values <1500 IU/litre AND conservative management criteria not met or not successful Offer as first line treatment (NICE 2019)
- BHCG level 1500 - 5000 IU/litre - Offer MTX or surgery (NICE 2019)
Haemodynamically stable
- Normal LFT's, U&E's and FBC
- Failed surgical treatment.
- Treatment should begin if the levels plateau
- If the levels are rising one must exclude intrauterine pregnancy before starting treatment
- Patient aware of possible lengthy follow-up process and is able to comply

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4.0 Exclusion Criteria

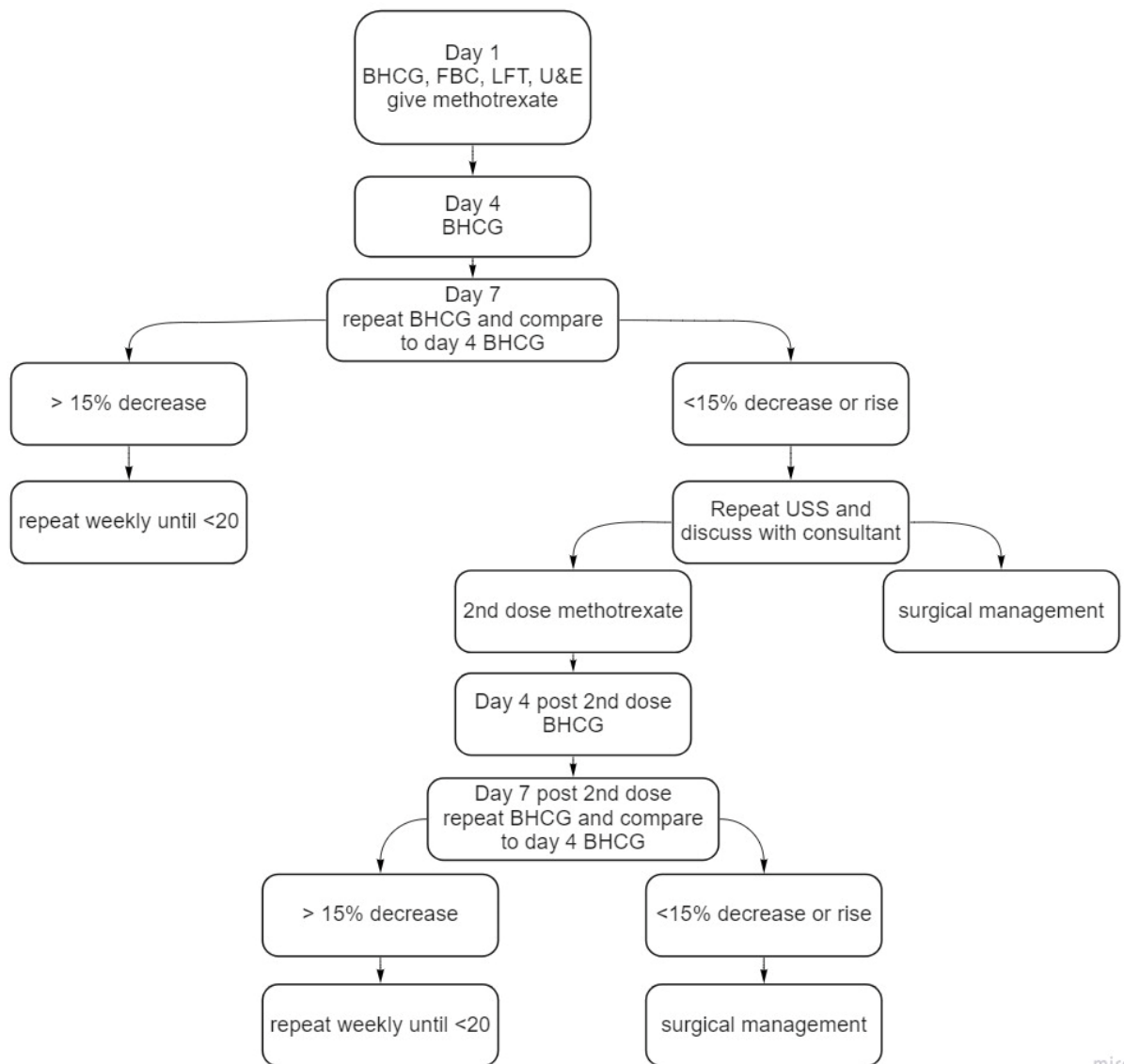
- The presence of cardiac activity in an ectopic pregnancy (live ectopic).
- Adnexal mass ≥ 35 mm diameter
- Evidence of intra-peritoneal haemorrhage (either clinically or on TVS)
- Active pulmonary disease
- Active infection or immuno-suppressed state including steroid use
- Peptic ulcer or ulcerative colitis
- Hepatic dysfunction, thrombocytopenia (platelet count <100), blood dyscrasia (WCC <2), severe anaemia
- Breastfeeding
- Difficulty or unwillingness of patient for prolonged follow-up (average follow-up 35 days).
- Patient unsure of choice of treatment

5.0 Treatment Protocol

- Registrar review and Consultant agreement – especially if the decision is made based on a single ultrasound scan
- Satisfy eligibility and exclusion criteria
- Discuss options for management surgical/medical
- Satisfy eligibility and exclusion criteria
- Counsel the patient and explain treatment protocol.
- Give information leaflet.
- Informed consent should be obtained and documented using eConsent, including unlicensed ('off label') use of methotrexate
- Record height and weight.
- Organise baseline blood tests: FBC, G&S, LFT's, U&E's and BHCG.
- Prescribe methotrexate $50\text{mg}/\text{m}^2$
- Methotrexate should be given as prescribed by appropriately trained staff in line with Trust Policy for administration of cytotoxic drugs.
- An appropriate area should be available for the patient to rest for up to one hour post administration of methotrexate. Before discharge home check patient has no local reaction to the drug.
- Before discharge ensure patient is fully aware of follow-up arrangements within the EPU

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- Do not prescribe anti-D for medical management of ectopic pregnancy (NICE2019)
- Patient should attend on Day 4 and Day 7 for BHCG; FBC
- If BHCG has fallen less than 15% of original value, between Day 4 and Day 7, a second dose of methotrexate should be considered after repeat ultrasound and discussion with on-call consultant.



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5.1 Outcome:

- 65.5 % to 94% successful treatment with single dose regime.
- Recurrent ectopic pregnancy rate 10 – 20%.
- Tubal patency approximately 80%.
- 10% patients require surgery for either ruptured ectopic pregnancy or the BHCG does not decline as expected.

BHCG levels are expected to rise initially up to Day 4 and then should decline by at least 15% between Day 4 and Day 7. If it does not decline as expected or plateaus, then consider second dose or surgery after discussion with consultant on-call. Trans-vaginal scan should be considered on Day 7 only if there is a suboptimal decline in BHCG between Days 4-7 or if the patient is symptomatic with worsening pain.

6.0 Information for Clinician

- Up to 75% of patients may complain of pain on days 3 – 7 (thought to be due to tubal miscarriage).
- BHCG levels may initially rise between days 1 – 4 (up to 86% of patients)
- The Mean time to resolution is 35 days
- A second dose of methotrexate may be given at Day 7 if the BHCG level fails to fall by more than 15% between days 4 – 7 (3-27% (in published literature), 14% of medically treated women will require more than one dose of methotrexate).
- Risk of tubal rupture is 7% and the risk remains while there is persistent BHCG.
- Avoid vaginal examination. TVS may be undertaken during first treatment week or subsequently if clinically indicated.

7.0 Information for Patients

- Medical treatment for ectopic pregnancy is now well established, and approximately 90% of patients do not require further surgery. Methotrexate is used for a variety of clinical conditions e.g. psoriasis, as well as for malignancies.
- It is important that all patients considering methotrexate understand that the companies manufacturing methotrexate have not licensed it ('off-label') to treat ectopic pregnancy, mainly because it was never anticipated for this use. This off-label treatment occurs fairly frequently (most medicines

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administered to children are off-label) in medicine. The treatment has been proven to work in selected patients and is recommended by NICE.

- Prolonged follow-up (up to 7 weeks) is required with blood tests until the BHCG level is below 20 u/L.
- A further dose of methotrexate may be necessary.
- Three quarters of women experience (worsening) abdominal pain following treatment, which is due to the drug acting on tubal pregnancy. It usually occurs on days 3-7 and normally lasts between 4-12 hours. However the patient should be advised to attend ED urgently if the pain is severe or there are other symptoms such as dizziness or shoulder tip pain.
- Some women will also experience some vaginal bleeding ranging from dark brown spotting to heavier bright red loss, which may last a few days to weeks. Advice is to contact EPU if it is excessive or the patient is concerned.
- Pregnancy should be avoided for 3 months after methotrexate has been given, because of a possible teratogenic effect – advice should be to use barrier contraception (RCOG)
- Side effects of the drug are minimal but may include nausea, vomiting, stomatitis and diarrhoea.
- Maintain ample fluid intake.
- Please avoid:
 - Alcohol or folic acid containing vitamins during treatment as they may interfere with the effectiveness of methotrexate treatment.
 - Sexual intercourse until resolution of the ectopic pregnancy.
 - Exposure to sunlight.
 - Heavy lifting.
- No difference in subsequent fertility rates between medical or surgical treatment

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8.0 References

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<https://www.nice.org.uk/guidance/ng126>

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Appendix A - Criteria for Assessment of Competence

Administration of Methotrexate Treatment for Patients undergoing Medical Management of Ectopic Pregnancy

Name:

Date:

Ward:

Clinical Services Unit:

Appointed staff will be able to demonstrate the knowledge, skills and competence necessary to safely administer Methotrexate treatment to patients undergoing Medical Management of Ectopic Pregnancy.

COMPETENCES	Signature	OUTCOME SAFE/UNSAFE
1. Understands professional accountability with regard to advice and information given to patients	Candidate	
	Assessor	
GENERAL		
1. Able to discuss local and national policy for safe handling and administration of cytotoxic drugs	Candidate	
	Assessor	
2. Able to discuss the management of spillage (and skin/eye/equipment contamination) of cytotoxic chemotherapy according to local guidelines	Candidate	
	Assessor	
3. Demonstrates competence in providing patient/carer education regarding all aspects of methotrexate treatment (selection and exclusion criteria, side effects, general advice etc.).	Candidate	
	Assessor	

PREPARATION AND ADMINISTRATION		
1. Able to discuss the procedures of ordering, transporting and storage of methotrexate relevant to own clinical area.	Candidate	
	Assessor	
2. Demonstrate awareness of the local policy for obtaining informed consent for administration of methotrexate.	Candidate	
	Assessor	
	Candidate	

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3. Demonstrates competence in pre-methotrexate treatment assessment (selection and exclusion criteria, blood tests, observations etc)	Assessor	
4. Able to discuss normal blood values and their relevance in the administration of methotrexate. Able to identify abnormal blood values relevant to administration of methotrexate and take appropriate action.	Candidate	
	Assessor	
5. Demonstrate knowledge of calculating drug doses and rate of administration.	Candidate	
	Assessor	
6. Interprets prescription correctly. Verifies methotrexate dosage, route and expiry date for individual patient. Checks details with competent second nurse/doctor prior to administration.	Candidate	
	Assessor	
7. Demonstrate competence in risk assessment in relation to handling, administration and disposal of cytotoxic drugs.	Candidate	
	Assessor	
8. Provides patient with appropriate written information regarding medical management of ectopic pregnancy and administration of methotrexate.	Candidate	
	Assessor	
9. Is able to give a full explanation of the procedure to the patient allowing time for questions. Maintains patient privacy and dignity at all times.	Candidate	
	Assessor	
10. Administers methotrexate according to practice guidelines and infection control policy wearing appropriate personal protection.	Candidate	
	Assessor	
11. Monitors patient and is able to identify signs of any adverse reaction during administration of methotrexate.	Candidate	
	Assessor	
12. Accurately records administration in all relevant documents.	Candidate	
	Assessor	
13. Disposes of equipment safely. Demonstrate the awareness of cytotoxic waste disposal and sharps disposal policy.	Candidate	
	Assessor	

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Appendix B - Guidelines for Assessors

The staff will be able to demonstrate the knowledge, skills and attitude necessary to undertake administration of methotrexate for patients undergoing medical management of ectopic pregnancy.

Criteria	Guidance
1. Understands professional accountability and responsibility	Candidate is able to clearly describe the role and practice to which this protocol relates, relevant codes of practice both professional and Trust wide, and own personal accountability related to the protocol.
2. Can discuss Consent Policy and Mental Capacity Act(2005)	Candidate is able to accurately identify and discuss key principles. Including demonstrating an awareness of the local policy for obtaining informed consent for the administration of methotrexate.
3. Demonstrates competence in relation to handling, administration and disposal of cytotoxic drugs.	Candidate is able to discuss both local and national policies regarding safe handling, administration and disposal of cytotoxic drugs. Can demonstrate understanding of calculating methotrexate dosages.
4. Is able to advise patient regarding all aspects of medical management of ectopic pregnancy.	Candidate can demonstrate full understanding of medical management of ectopic pregnancy, including selection and exclusion criteria, side effects of methotrexate, follow up and aftercare. In line with local protocol.
5. Provides appropriate written information to patient.	Candidate gives patient appropriate written information regarding treatment.
6. Is able to give full explanation of the procedure prior to administration of methotrexate, allowing time for questions.	Candidate is able to discuss procedure with patient and can identify issues that require further explanation.
7. Is able to interpret prescription correctly and check details in line with local and national policies.	Candidate can demonstrate correct interpretation of prescription and check all details in line with Trust and NMC Standards
8. Is able to complete all relevant documentation appropriately.	Candidate demonstrates that documentation is completed accurately and legibly and in an appropriate time frame.

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Appendix C - Assessment Summary

ASSESSMENT SUMMARY FOR

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This is to certify thathas successfully
completed the competencies related to **Administration of
Methotrexate for patients undergoing Medical Management of
Ectopic Pregnancy.**

Signed.....

Name (Print).....

Position

Date.....

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